

## LJUNGBY, Sweden

WMO: 026220

Lat: 56.8525N

Lon: 13.8794E

Elev: 486

StdP: 14.44

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | 3.2        | 9.6 | -0.3                          | 5.5 | 5.1  | 5.7 | 7.5 | 12.0 | 18.8                  | 38.6 | 16.9 | 37.5 | 0.9                      | 10   | 0.567 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 20.9                   | 80.0            | 65.2 | 76.9 | 64.0 | 73.8 | 62.3 | 67.6                | 75.6 | 65.7 | 73.4 | 64.0 | 71.2 | 6.0                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 64.8                            | 93.6 | 71.2 | 62.9 | 87.2 | 68.8 | 61.1 | 81.8 | 67.2 | 32.3          | 75.3 | 30.8 | 73.9 | 29.5 | 71.4 | 74.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 14.6          | 13.2 | 11.8 | -3.0                       | 84.0 | 8.2                | 4.0 | -8.9                                               | 86.9 | -13.7      | 89.2 | -18.3      | 91.5 | -24.3      | 94.4 |    |    |
| (5)               |      |      | -4.0                       | 70.8 | 8.0                | 2.4 | -9.8                                               | 72.5 | -14.5      | 73.9 | -19.0      | 75.2 | -24.8      | 76.9 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 44.1   | 28.4  | 29.5  | 33.7  | 42.2  | 50.5  | 56.4  | 60.8  | 59.1  | 52.5  | 44.5  | 37.8  | 32.3  |  |  |
|                                                                        | DBStd       | 13.09  | 8.79  | 7.72  | 7.58  | 5.98  | 6.43  | 5.29  | 4.86  | 4.71  | 5.19  | 6.70  | 6.83  | 8.82  |  |  |
|                                                                        | HDD50       | 3212   | 669   | 574   | 505   | 243   | 70    | 8     | 0     | 1     | 35    | 193   | 366   | 548   |  |  |
|                                                                        | HDD65       | 7674   | 1134  | 994   | 969   | 684   | 450   | 263   | 149   | 190   | 376   | 637   | 815   | 1013  |  |  |
|                                                                        | CDD50       | 1046   | 0     | 0     | 0     | 9     | 87    | 199   | 335   | 284   | 109   | 22    | 1     | 0     |  |  |
|                                                                        | CDD65       | 33     | 0     | 0     | 0     | 0     | 1     | 5     | 19    | 8     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 628    | 0     | 0     | 0     | 0     | 32    | 150   | 340   | 101   | 5     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 85     | 0     | 0     | 0     | 0     | 1     | 18    | 56    | 10    | 0     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 5.4    | 6.2   | 6.1   | 5.9   | 5.5   | 5.3   | 4.9   | 4.5   | 4.3   | 4.8   | 5.2   | 6.0   | 5.9   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 32.0   | 2.6   | 2.0   | 1.7   | 1.8   | 2.3   | 2.8   | 3.3   | 3.3   | 2.9   | 3.1   | 3.1   | 2.8   |  |  |
|                                                                        | PrecMax     | 41.0   | 5.9   | 5.3   | 4.5   | 3.7   | 4.9   | 5.9   | 8.2   | 5.5   | 6.8   | 6.5   | 6.1   | 6.5   |  |  |
|                                                                        | PrecMin     | 23.0   | 0.2   | 0.0   | 0.4   | 0.4   | 0.4   | 0.0   | 0.4   | 0.6   | 0.5   | 0.4   | 1.5   | 0.8   |  |  |
|                                                                        | PrecStd     | 4.6    | 1.5   | 1.2   | 0.9   | 0.9   | 1.1   | 1.3   | 1.9   | 1.4   | 1.6   | 1.4   | 1.1   | 1.3   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 45.9  | 48.8  | 58.1  | 69.9  | 77.3  | 82.5  | 85.3  | 80.8  | 73.1  | 62.7  | 53.0  | 49.3  |  |  |
|                                                                        |             | MCWB   | 44.9  | 44.3  | 47.1  | 55.0  | 60.7  | 63.5  | 66.9  | 67.6  | 62.3  | 56.8  | 51.6  | 47.9  |  |  |
|                                                                        | 2%          | DB     | 43.1  | 44.0  | 52.5  | 64.3  | 73.9  | 78.4  | 81.0  | 76.9  | 68.6  | 58.2  | 50.7  | 46.9  |  |  |
|                                                                        |             | MCWB   | 42.0  | 42.0  | 44.5  | 51.0  | 59.3  | 63.3  | 66.0  | 65.2  | 60.5  | 55.2  | 49.4  | 45.7  |  |  |
|                                                                        | 5%          | DB     | 41.1  | 41.8  | 47.2  | 59.7  | 70.0  | 74.4  | 78.0  | 73.3  | 65.5  | 56.1  | 48.8  | 44.7  |  |  |
|                                                                        |             | MCWB   | 40.1  | 40.2  | 42.1  | 48.7  | 56.4  | 61.3  | 65.0  | 63.5  | 58.7  | 53.6  | 47.7  | 43.4  |  |  |
|                                                                        | 10%         | DB     | 39.4  | 39.6  | 43.9  | 55.5  | 65.1  | 70.3  | 74.5  | 70.2  | 62.4  | 54.1  | 47.0  | 42.9  |  |  |
|                                                                        |             | MCWB   | 38.5  | 38.3  | 40.6  | 46.5  | 54.4  | 59.0  | 63.6  | 62.4  | 57.0  | 51.9  | 46.0  | 41.9  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 45.1  | 45.5  | 48.8  | 55.6  | 64.2  | 68.5  | 70.5  | 69.7  | 64.5  | 57.9  | 52.2  | 48.9  |  |  |
|                                                                        |             | MCDB   | 45.6  | 47.6  | 55.5  | 66.9  | 74.3  | 77.1  | 79.8  | 76.9  | 70.5  | 60.4  | 52.7  | 49.6  |  |  |
|                                                                        | 2%          | WB     | 42.2  | 42.6  | 45.4  | 52.7  | 61.5  | 64.6  | 68.2  | 67.0  | 61.7  | 55.9  | 49.7  | 46.0  |  |  |
|                                                                        |             | MCDB   | 42.7  | 43.7  | 50.3  | 62.3  | 70.1  | 74.8  | 76.7  | 73.4  | 67.0  | 57.9  | 50.4  | 46.7  |  |  |
|                                                                        | 5%          | WB     | 40.2  | 40.2  | 43.1  | 49.9  | 58.5  | 62.4  | 66.5  | 64.9  | 59.3  | 54.1  | 47.8  | 43.7  |  |  |
|                                                                        |             | MCDB   | 40.9  | 41.4  | 46.4  | 56.4  | 66.0  | 71.7  | 74.9  | 70.8  | 64.2  | 55.7  | 48.6  | 44.4  |  |  |
|                                                                        | 10%         | WB     | 38.5  | 38.3  | 41.0  | 47.5  | 55.8  | 60.2  | 64.7  | 63.3  | 57.5  | 52.3  | 46.2  | 41.9  |  |  |
|                                                                        |             | MCDB   | 39.2  | 39.6  | 43.5  | 54.2  | 63.4  | 68.4  | 72.6  | 68.8  | 61.5  | 53.8  | 46.9  | 42.8  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 9.0   | 10.3  | 15.1  | 20.8  | 21.7  | 22.0  | 20.9  | 19.4  | 17.8  | 13.5  | 8.5   | 8.2   |  |  |
|                                                                        |             | MCDBR  | 7.7   | 11.4  | 21.0  | 30.2  | 30.0  | 31.3  | 29.4  | 25.0  | 23.0  | 14.5  | 8.4   | 7.7   |  |  |
|                                                                        |             | MCWBR  | 7.4   | 9.1   | 14.2  | 18.5  | 16.7  | 17.4  | 16.1  | 15.1  | 15.4  | 11.2  | 7.9   | 7.4   |  |  |
|                                                                        | 5% WB       | MCDBR  | 7.9   | 9.9   | 18.3  | 25.1  | 24.2  | 25.9  | 25.2  | 21.1  | 20.5  | 12.9  | 8.4   | 7.9   |  |  |
|                                                                        |             | MCWBR  | 7.8   | 8.3   | 13.1  | 16.4  | 14.7  | 15.8  | 15.3  | 13.7  | 14.4  | 10.6  | 7.9   | 7.8   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.294 | 0.322 | 0.342 | 0.378 | 0.376 | 0.367 | 0.389 | 0.388 | 0.357 | 0.345 | 0.321 | 0.284 |  |  |
|                                                                        | taud        |        | 2.273 | 2.315 | 2.348 | 2.319 | 2.384 | 2.432 | 2.384 | 2.391 | 2.465 | 2.419 | 2.340 | 2.240 |  |  |
|                                                                        | Ebn at Noon |        | 187   | 231   | 258   | 265   | 274   | 278   | 268   | 260   | 252   | 221   | 175   | 155   |  |  |
|                                                                        | Edh at Noon |        | 16    | 23    | 29    | 35    | 35    | 34    | 35    | 32    | 26    | 21    | 15    | 13    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 127   | 317   | 762   | 1251  | 1636  | 1736  | 1596  | 1282  | 893   | 421   | 158   | 88    |  |  |
|                                                                        | RadStd      |        | 24    | 53    | 78    | 169   | 160   | 129   | 199   | 120   | 109   | 70    | 30    | 9     |  |  |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (65 neighbors) |       | +0.74   | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | -286  | N/A   | N/A   |

Nomenclature: See separate page